

Project FORESIGHT

Demand & Inventory Intelligence using Machine Learning

1. Project Overview

Project FORESIGHT is a Machine Learning-based Demand Forecasting and Inventory Intelligence System designed to help retail and supply chain businesses accurately predict future product demand and optimize inventory levels.

The system analyzes historical sales, inventory snapshots, product information, and calendar events to generate demand forecasts and identify stockout and overstock risks. An interactive Streamlit dashboard provides business insights and supports data-driven decision-making.

2. Problem Statement

Retail businesses often struggle with inaccurate demand estimation, resulting in either excess inventory or stock shortages.

These challenges lead to:

- Revenue loss
- Increased warehouse costs
- Poor customer satisfaction
- Supply chain inefficiencies

The objective of this project is to build a predictive system that forecasts future demand and recommends inventory actions to minimize these problems.

3. Dataset Description

The project uses four datasets.

1. sales_daily.csv

Column	Description
date	Sales Date
sku_id	Product ID
units_sold	Quantity Sold
unit_price	Selling Price
promotion	Promotion Status

2. sku_master.csv

Column	Description
sku_id	Product ID
product_name	Product Name
category	Product Category
list_price	MRP
reorder_point	Minimum Inventory Level

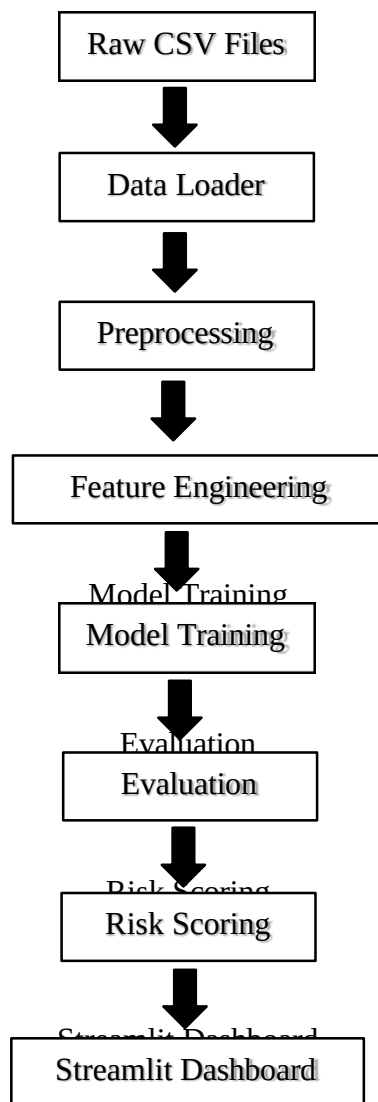
3. calendar.csv

Column	Description
date	Calendar Date
day_name	Day Name
month	Month
is_weekend	Weekend Flag
is_holiday	Holiday Flag

4. inventory_snapshots.csv

Column	Description
date	Inventory Date
sku_id	Product ID
on_hand_units	Current Stock
on_order_units	Incoming Stock

4. System Architecture



5. Exploratory Data Analysis (EDA)

EDA is performed to understand the data before training the model.

Activities include:

- Missing Value Analysis
- Duplicate Detection
- Statistical Summary
- Sales Trend Analysis
- Monthly Sales Trend
- Category-wise Sales
- Top Selling Products

- Inventory Analysis
- Correlation Analysis

Key observations:

- Sales increase during weekends and holidays.
- Furniture category contributes the highest revenue.
- Promotional products show higher demand.
- Some SKUs consistently remain understocked.

6. Feature Engineering

The following features are created for model training:

Date Features

- Year
- Month
- Week
- Day
- Day of Week
- Quarter
- Weekend Indicator

Sales Features

- Lag 1
- Lag 7
- Lag 14
- Rolling Mean (7 Days)
- Rolling Mean (30 Days)
- Rolling Standard Deviation

Inventory Features

- Inventory Gap
- Total Inventory
- Reorder Point Difference

Pricing Features

- Discount Percentage
- Price Difference

These engineered features significantly improve model performance.

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8. Results

Model performance is evaluated using regression metrics.

Metric	Example Value
MAE	18.42
RMSE	24.16
MAPE	7.21%
WAPE	6.84%
R ² Score	0.94

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Risk analysis identifies products with high stockout and overstock probability, enabling proactive inventory management.

10. Conclusion

Project FORESIGHT demonstrates how Machine Learning can improve demand forecasting and inventory management.

The system enables businesses to:

- Forecast future product demand accurately
- Reduce stockout and overstock risks
- Improve inventory planning
- Lower operational costs
- Enhance customer satisfaction through better product availability

The solution is modular, scalable, and suitable for integration into retail analytics platforms.